

DIABETES MELLITUS

The body cannot make insulin, cannot use it well, or both — so glucose stays in the **blood** instead of entering the **cells**.

Type 1

NO insulin · autoimmune destruction of beta cells

- **Absolute** insulin deficiency — **insulin is required** for life
- Often diagnosed young, but can occur at any age
- **Rapid onset**; often thin or weight loss at diagnosis
- Crisis risk: **DKA**

Type 2

RESISTANCE to insulin · **plus relative deficiency**

- Cells respond poorly to insulin; production falls over time
- Most common type; risk rises with age, excess weight, inactivity, family history
- **Gradual onset**; may have no symptoms for years
- Lifestyle, oral or injectable meds; may need insulin. Crisis risk: **HHS**

HYPOGLYCEMIA · BELOW 70 mg/dL

"Cold and clammy — need some candy"

- Shaky, **sweaty**, pale, cool clammy skin
- Tachycardia, anxiety, hunger, irritability
- Confusion, slurred speech → seizure, loss of consciousness
- **Rule of 15**: 15 g fast-acting carbohydrate → recheck in 15 min → repeat if still under 70. Then a snack or meal
- **Unable to swallow safely**: glucagon or IV dextrose per protocol — nothing by mouth

HYPERGLYCEMIA

"Hot and dry — sugar high"

- **3 Ps**: **polyuria**, **polydipsia**, **polyphagia**
- Warm, dry skin; dehydration
- Blurred vision, fatigue, headache
- Slow wound healing, frequent infections
- Untreated → **DKA** (type 1) or **HHS** (type 2)

DIAGNOSIS (ADA)

A1C	≥ 6.5%
Fasting glucose	≥ 126 mg/dL
2-h OGTT	≥ 200 mg/dL
Random + symptoms	≥ 200 mg/dL
Prediabetes A1C	5.7–6.4%

A1C reflects average glucose over about 3 months.

INSULIN — ONSET, PEAK, DURATION approximate; varies by product — confirm in a drug reference

Type	Examples	Onset	Peak	Duration	Nursing watch-for
Rapid-acting	lispro, aspart, glulisine	~15 min	1–2 h	3–5 h	Give with the meal — food must be available
Short-acting	regular	30–60 min	2–4 h	5–8 h	The insulin given IV (DKA, HHS)
Intermediate	NPH	1–2 h	4–12 h	12–18 h	Cloudy — roll, don't shake. Mixing with regular: clear before cloudy
Long-acting	glargine, detemir, degludec	1–2 h	no peak	~24 h+	Basal insulin — never mix with other insulins

Hypoglycemia risk is highest at the peak. Insulin is a high-alert medication — follow double-check policy.

DKA VS HHS

	DKA — diabetic ketoacidosis	HHS — hyperosmolar hyperglycemic state
Usually	Type 1	Type 2, often older adults
Glucose	Elevated, typically > 250 mg/dL	Very high, typically > 600 mg/dL
Ketones / pH	Ketones present, acidosis — pH < 7.3	Minimal or no ketones; pH > 7.3
Signs	Kussmaul respirations, fruity breath , abdominal pain, N/V	Severe dehydration , high osmolality, altered mental status
Onset	Hours to a day	Days
Priorities	IV fluids first → IV regular insulin as ordered → monitor K⁺ closely (insulin drives K ⁺ into cells → hypokalemia) → hourly glucose, I&O, neuro checks	

NURSING PRIORITIES

- Check glucose before meals and at bedtime, or as ordered
- Coordinate insulin with meal trays
- Rotate injection sites within one area
- Watch for hypoglycemia when NPO, vomiting, or at insulin peak

SICK-DAY RULES

- **Do not stop insulin** when ill, even if not eating
- Check glucose more often
- Check ketones, especially type 1
- Fluids to prevent dehydration; report persistent vomiting

FOOT CARE TEACHING

- Inspect feet daily; use a mirror
- **Never go barefoot**; well-fitting shoes
- Wash and dry well, especially between toes
- Report cuts, blisters, color change